

	Title	Data transmission protocol
	File	BM77_transmissionprotocol_rev01_20190807

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1. USB

1.1 General Information

The blood pressure monitor BM77 is able to transfer stored measurement data via USB. The device is a HID device, so no driver installation is necessary. The Vendor ID (VID) is: 0x2828 and the Product ID (PID) is: 0x9001.

1.2 Connection

This command starts a connection between the PC and the BM77.

Name	Format	Info	Note
Command	1 byte	The command byte is 0x5A	
Filler bytes	7 byte	The following bytes are filler bytes with 0x00	

Example:

Sending: [0x5A 0x00 0x00 0x00 0x00 0x00 0x00 0x00]

Meaning: 0x5A – Command byte | 7 bytes with 0x00 – Filler bytes

The response from the device to the connection attempt.

Name	Format	Info	Note
Length of Payload	1 byte	Value of 0x04 means payload is 4 byte	
Response	1 byte	The response for the connection attempt is 0x5B	
Amount of measurments of User 1	1 byte	The value of one equals one measurement for user 1	
Amount of measurments of User 2	1 byte	The value of one equals one measurement for user 2	
Start of first measurement	1 byte	Depending the on the length of the payload the first measurement starts here or in the next package	
Filler bytes	3 byte	Filler bytes always 0x00	

Example:

Receiving: [0x04 0x5B 0x05 0x04 0xXX 0x00 0x00 0x00]

Meaning: 0x04 – Payload is 4 byte | 0x5B – Command Byte | 0x05 – 5 measurements for user 1 | 0x04 – 4 measurements for user 2 | 0xXX depending on Payload first byte of measurement | 3 bytes with 0x00 – Filler bytes

1.3 Measurements

After the connection is started the BM77 sends the measurements.

The first transmitted measurement is the newest measurement of user 1; the last transmitted measurement is the oldest measurement of user 2. The first byte of each package shows the length of the payload. The following table does not show the values of one package it shows how the bytes have to be interpreted depending on the payload of the package. One measurement has 8 byte in total For example: The first byte of the first package is 0x05, which means the next 5 bytes are the payload. The next package has a payload of 0x06. This means the first measurement uses 5 bytes of the first package and 3 of the second package. The second measurement uses the remaining 3 bytes of the second package.

Name	Format	Info	Note
Year/Month	1 byte	The value of bit 0-3 equals the month The value of bit 4-7 equals the low bits of the year	
Year/Day	1 byte	The value of bit 0-4 equals the day The value of bit 5-7 equals the high bits of the year	
Hour	1 byte	The value equals the hour of the measurement	
Minute	1 byte	The value equals the minute of the measurement	

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IHB/HSD/ Systolic	1 byte	The value of bit 0 equals the high bit of the systolic bit 1-4 are filler bits bit 5 and bit 6 shows the HSD value (MSB LSB): 00 = HSD not detected , 01 =HSD detected , 10 = unable to judge MSO to LSO bit 7 shows the Irregular heart beat 0 means no IHB and 1 means IHB detected	
Systolic	1 byte	The value equals the low bits of the systolic	
Diastolic	1 byte	The value equals the diastolic	
Pulse	1 byte	The value equals the pulse	

Example:

Receiving: [0x37 0x26 0x0A 0x06 0xA0 0x78 0x50 0x3C]

Meaning: 0x37 26 - 6th of July2019 | 0x0A – Hour 10 | 0x06 – Minute 6 | 0xA0 78 – HSD detected, IHB detected

Systole: 120mmHg | 0x50 – Diastole:80mmHg | 0x3C – Pulse: 60bpm

After the last measurement one package with 8 filler bytes with the value of 0x00 is transmitted and then one package with the Checksum. The first byte of the last package is the length of the payload of the checksum (normally 2); the next two bytes are the checksum and is calculated as follows: All 8 bytes of each measurement are added.

1.4 End current communication

If the checksum is correct PC sends:

Name	Format	Info	Note
Command	1 byte	The command to end the communication is 0x5A	
Filler bytes	7 byte	7 filler bytes with the value of 0x00	

Example:

Sending: [0x5A 0x00 0x00 0x00 0x00 0x00 0x00 0x00]

Meaning: 0x5A – Command byte | 7 bytes with 0x00 – Filler bytes

2. Bluetooth Low energy

2.1 General Information

The BM77 is able to transfer stored measurement data via Bluetooth Low energy. The BM77 always works as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet.

2.2 Advertising Data

The advertising name is “BM77”

2.3 Services and Characteristics

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Blood Pressure	0x1810	00001810-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb	
Battery Service	0x180F	0000180F-0000-1000-8000-00805f9b34fb	Not used!
Health Thermometer	0x1809	00001809-0000-1000-8000-00805f9b34fb	Not used!
Unknown Service		0000F880-0000-1000-8000-00805f9b34fb	Not used!

Generic Access

Name	Assigned Number	Properties	Note
Device Name	0x2A00	Read, Write	“Beurer BM77”
Appearance	0x2A01	Read	“0” - Unknown

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Peripheral Privacy Flag	0x2A02	Read, Write	"0x00"
Reconnection Address	0x2A03	Read, Write	
Peripheral Preferred Connection Parameters	0x2A04	Read	Connection Interval: 30,00ms – 40,00ms Slave Latency: 0 Supervision Timeout Multiplier: 3200

Blood Pressure

Name	UUID	Properties	Note
Blood Pressure Measurement	0x2A35	Indicate	
Intermediate Cuff Pressure	0x2A36	Notify	Not used by BM77
Blood Pressure Feature	0x2A49	Read	No features supported

Device Information

Name	Assigned Number	Properties	Note
Manufacturer Name String	0x2A29	Read	"BEURER"
Model Number String	0x2A24	Read	"BT A8105"
Serial Number String	0x2A25	Read	"00001"
Hardware Revision String	0x2A27	Read	"A810501"
Firmware Revision String	0x2A26	Read	"160329"
Software Revision String	0x2A28	Read	"1000"

2.4 Blood Pressure Measurement

The BM77 is sending the data automatically after a connection has been established successfully. This can be triggered by clicking on the M1 or M2 short after starting the device. It is also triggered after every measurement. The device is sending all data from both users.

Blood Pressure Measurement – Indicate

For every measurement stored on the BM77 an indication is send.

Name	Format	Info	Note
flags	1 byte	See separate Flag table "flag description". Value is always 0x1E.	
systolic	sfloat (2byte)	The value of 1 equals 1,0 mmHg. The valid range is 0-300.	
diastolic	sfloat (2byte)	The value of 1 equals 1,0 mmHg. The valid range is 0-300.	
mean arterial pressure	sfloat (2byte)	Not supported by the BM77. The value is always 0.	
timestamp	time (7byte)	See table "Timestamp description" for the timestamp data	
pulse rate	sfloat (2byte)	The value of 1 equals a pulse of 1. The valid range is 0-255.	
user ID	uint8 (1byte)	The value of 0 represents measurement from user 1 and the value of 1 measurements of user 2	
measurement status	2 byte	See separate table "measurement status description".	

Example

Receiving: [0x1E 0x70 0x00 0x4D 0x00 0x00 0x00 0xDF 0x07 0x01 0x0E 0x0A 0x37 0x00 0x48 0x00 0x00 0x40 0x00]

Meaning: 0x56 – The flags set for the BM77 | 0x00 0x70 – Systolic value of 117 | 0x00 0x4D – Diastolic value of 77 | 0x00 0x00 – MAP value, not supported on the BM77 and always 0 | 0x07 0xDF – Year 2015 | 0x01 – Month January | 0x0E – Day 14 | 0x0A – Hour 10 | 0x37 – Minute 55 | 0x00 Second 0 | 0x48 0x00 – Pulse of 72 | 0x00 – User 1 | 0x40 00 - Measurement status HSD detected.

Flag description – Part of Blood Pressure Measurement

The flags have a size of 1 byte and are part of the blood pressure measurement Indication. The value for the BM77 is always 0x1E.

Name	Format	Info	Note
unit flag	1 bit	systolic, diastolic and MAP unit: 0 = mmHg 1 = kPa	
time stamp flag	1 bit	0 = not present 1 = present	
pulse rate flag	1 bit	0 = not present 1 = present	
user ID flag	1 bit	0 = not present 1 = present	
measurement status flag	1 bit	0 = not present 1 = present	
reserved for future use	1 bit		
HSD flag	1 bit	0 = not present 1 = present	
reserved for future use	1 bit		

Timestamp description – Part of Blood Pressure Measurement

The timestamp has a size of 7 bytes and is part of the blood pressure measurement Indication.

Name	Format	Info	Note
year	uint16 (2 byte)	The value of 1 equals 1 year. The valid range is from 1582-9999	
month	uint8 (1 byte)	The valid range is from 0-12. The value 0 represents unknown, the value of 1 represents January, the value of 2 represents February ...	
day	uint8 (1 byte)	The valid range is from 0-31. The value 0 represents unknown, the value of 1 represents the first day, the value of 2 represents the second day...	
hours	uint8 (1 byte)	The value of 1 equals 1 hour. The valid range is from 0-23	
minutes	uint8 (1 byte)	The value of 1 equals 1 minute. The valid range is from 0-59	
seconds	uint8 (1 byte)	The value of 1 equals 1 second. The valid range is from 0-59	

Measurement Status – Part of Blood Pressure Measurement

The measurement status has a size of 2 bytes and is part of the blood pressure measurement Indication. The value of the measurement status is reversed the order is MSO to LSO.

Name	Format	Info	Note
body movement	1 bit	0 = no body movement 1 = body movement	
cuff fit	1 bit	0 = cuff fits properly 1 = cuff is too loose	
irregular pulse	1 bit	0 = no irregular pulse 1 = irregular pulse	
pulse rate range	2 bit	00 = in range 01 = exceeds upper limit 10 = below lower limit 11 = reserved for future use MSO to LSO	
measurement position	1 bit	0 = proper position 1 = improper position	
HSD	2 bit	00 = HSD not detected 01 =HSD detected 10 = unable to judge MSO to LSO	Not defined by Bluetooth SIG. The HSD value is represented by 2 bit.
reserved for future use	8 bit		

2.5 Blood Pressure Feature

Blood Pressure Feature – Read

The blood pressure feature tells the user which features are supported. The value for the BM77 is always [0x00 00] but the device supports Irregular Heart Beat and HSD.

Name	Format	Info	Note
body movement feature	1 bit	0 = feature not supported 1 = feature supported	
cuff fit feature	1 bit	0 = feature not supported 1 = feature supported	
irregular pulse feature	1 bit	0 = feature not supported 1 = feature supported	
pulse rate range	1 bit	0 = feature not supported 1 = feature supported	
measurement position	1 bit	0 = feature not supported 1 = feature supported	

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multiple bond feature	1 bit	0 = feature not supported 1 = feature supported	
HSD feature	1 bit	0 = feature not supported 1 = feature supported	
reserved for future use	9 bit		

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